

Cognitive Underwater Acoustic Networking and Target Tracking: A Comprehensive Survey

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Abstract—Underwater acoustic networks are evolving from static, manually-configured systems into cognitive, learning-enabled platforms that can perceive, reason, and adapt to harsh ocean dynamics in real-time. Accurate target tracking is a core service of these networks and underpins marine resource exploration, environmental monitoring, and maritime security. Existing reviews or surveys, however, rarely examine underwater acoustic target tracking through the lens of cognitive communications and networking, and often offer a narrow perspective on addressing the paradigm shifts driven by emerging technologies like deep learning. To fill this gap, this work presents a systematic survey of this field and introduces an innovative three-dimensional taxonomy framework based on the three levels of the cognitive underwater acoustic target tracking network: the target layer, the perception layer, and the processing layer. Within this framework, we comprehensively survey the literature

over the period 2016-2025, spanning from the theoretical foundations to diverse algorithmic approaches in underwater acoustic target tracking. Particularly, we emphasize the transformative potential and recent advancements of machine learning techniques, including deep learning and reinforcement learning, in enhancing the performance and adaptability of cognitive underwater tracking systems. Finally, this survey concludes by identifying key challenges in the field and proposing future avenues based on emerging technologies such as data desensitization, embodied intelligence, and large models.

Index Terms—Underwater acoustics, target tracking, cognitive system, state estimation, deep learning.

I. INTRODUCTION

THE ocean harbors vast reserves of resources and energy, serving not only as a vital repository but also as another crucial domain for human sustainable development, drawing increasing global attention in recent decades. Simultaneously, marine ecosystems provide significant ecological services to humanity, including gas regulation, nutrient cycling, and waste management. Moreover, ocean governance underpins national security and sovereignty [1]. Consequently, perceiving and understanding the unknown marine environment, effectively utilizing oceanic resources, strengthening the protection of marine ecosystems, and safeguarding national maritime rights have become international consensus [2].

The accurate and effective tracking of underwater targets is fundamental to maritime security, resource management, and environmental protection. In this context, cognitive networks leveraging advanced sensors have become a key area of research and development. Progress in this field is being propelled by recent innovations in sensor technology, new paradigms in cognitive communication, and the transformative impact of artificial intelligence. Together, these developments are enabling sophisticated applications in critical sectors, including defense, environmental surveillance, and marine exploration [3], [4].

Due to the complexity of the marine environment and the diversity of targets, the development of underwater target tracking technology faces significant challenges [5]. The propagation characteristics of signals are significantly influenced by marine conditions [6], and underwater targets typically exhibit high maneuverability and stealth capabilities [7], making them difficult to accurately perceive. Traditional terrestrial tracking media, including electromagnetic waves and lasers, experience severe absorption and attenuation in seawater,

Received 9 December 2025; accepted 22 January 2026. Date of publication 28 January 2026; date of current version 9 February 2026. This work was supported by the National Natural Science Foundation of China under Grant 72501291 and in part by the Youth Independent Innovation Foundation of NUDT under Grant ZK-2023-21. The associate editor coordinating the review of this article and approving it for publication was D. Niyato. (Zhong Yang and Zhengqiu Zhu are co-first authors.) (Corresponding authors: Zhengqiu Zhu; Guangquan Cheng; Jun Zhang.)

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Digital Object Identifier 10.1109/TCCN.2026.3658820

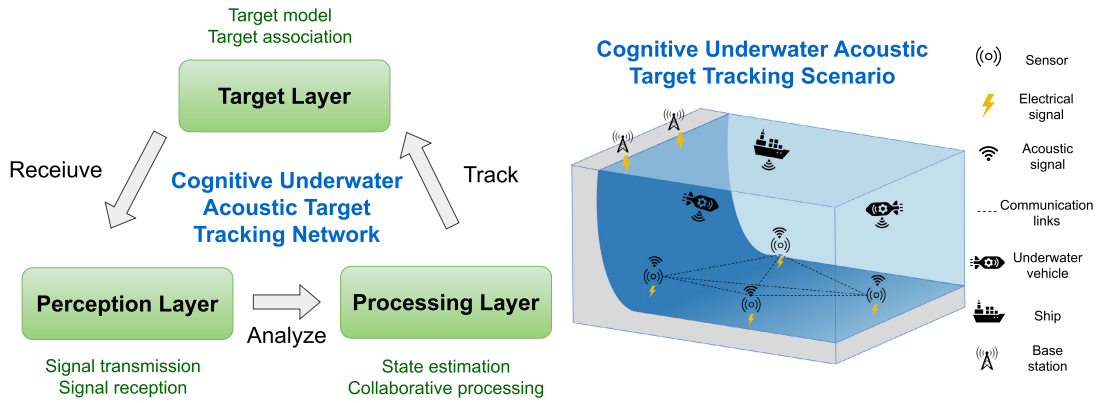


Fig. 1. Cognitive underwater acoustic target tracking network structure and scenario. The left panel depicts the hierarchical architecture of the cognitive network, comprising the target layer, the perception layer, and the processing layer. These layers are interconnected to establish a closed loop. The right panel illustrates the scenario of underwater acoustic target cognitive tracking, highlighting key elements including acoustic targets, sensors, and their interrelationships.

resulting in poor performance in marine applications [8]. In contrast, sound waves are the only medium that can still propagate over long distances underwater, enabling accurate cognition of underwater targets.

A. Definition and Structural Division of Cognitive Network

Cognition is defined as the capability of a network to infer environmental and target states within resource-constrained underwater environments by utilizing both prior knowledge and measurement data. Operating on a perception-action closed loop, the network proactively adjusts observation and transmission strategies to minimize uncertainty regarding the target state. The core philosophy of cognitive networks-perceiving the environment through interaction and dynamically adapting parameters to meet end-to-end objectives-is inherently well-suited to address the complexities of underwater acoustic tracking. Drawing upon Cognitive Dynamic System (CDS) theory [9], the operation of an underwater acoustic target tracking cognitive network is modeled as a closed-loop process driven by a perception-action cycle. From a systems theory perspective, any detection system, whether active or passive, comprises three fundamental entities: the object within the environment, the interface for data acquisition, and the processing subject [10].

Consequently, we structure the underwater acoustic target tracking cognitive network into three distinct layers: the **target layer**, the **perception layer**, and the **processing layer**. This hierarchical architecture effectively decouples the observed object, the observation mechanism, and the cognitive decision-making center. Furthermore, it accommodates the distinct stages of the perception-action cycle, thereby facilitating robust closed-loop control [11]. Figure 1 illustrates the structure of the cognitive underwater acoustic target tracking network and the problem scenario.

B. Motivation for Building a Multidimensional Taxonomy Framework

Due to the importance of underwater acoustic target tracking in military and civilian fields, researchers have conducted comprehensive investigations into these technologies [12], [13],

[14], [15], [16]. In addition, the integration of artificial intelligence, big data analysis, and embodied intelligence has created unprecedented opportunities for technological breakthroughs in this field, making it a key driver of future innovation in marine science. However, existing reviews or surveys in this field have two significant limitations [17]:

- First, cognitive underwater acoustic target tracking encompasses multiple research dimensions, but current reviews typically adopt narrow perspectives and scenario frameworks. Consequently, these reviews have limited applicability in different operational environments, thereby hindering readers and researchers from forming a systematic understanding of the tracking cognitive process.
- Second, most existing reviews were published prior to the paradigm shift driven by recent advancements in deep learning, including the application of deep neural networks in measurement data analysis and the application of deep reinforcement learning in sensor control coordination, both of which provide significant optimization across the three layers of cognitive underwater acoustic target tracking networks. Consequently, these reviews lack coverage of cutting-edge developments in cognitive technologies, which are fundamentally transforming the methodological approaches in this field.

This evident research gap underscores the urgent need for a comprehensive survey that systematically synthesizes both fundamental methodologies and contemporary innovations in underwater acoustic target tracking.

C. Scope of the Survey

As illustrated in Figure 2, cognitive underwater acoustic target tracking has attracted increasing research attention in recent years, with a large number of publications emerging in this field.

This survey systematically examines the problem scenarios of cognitive underwater acoustic target tracking, analyzes the key components of cognitive systems, and proposes a comprehensive three-dimensional classification framework based on

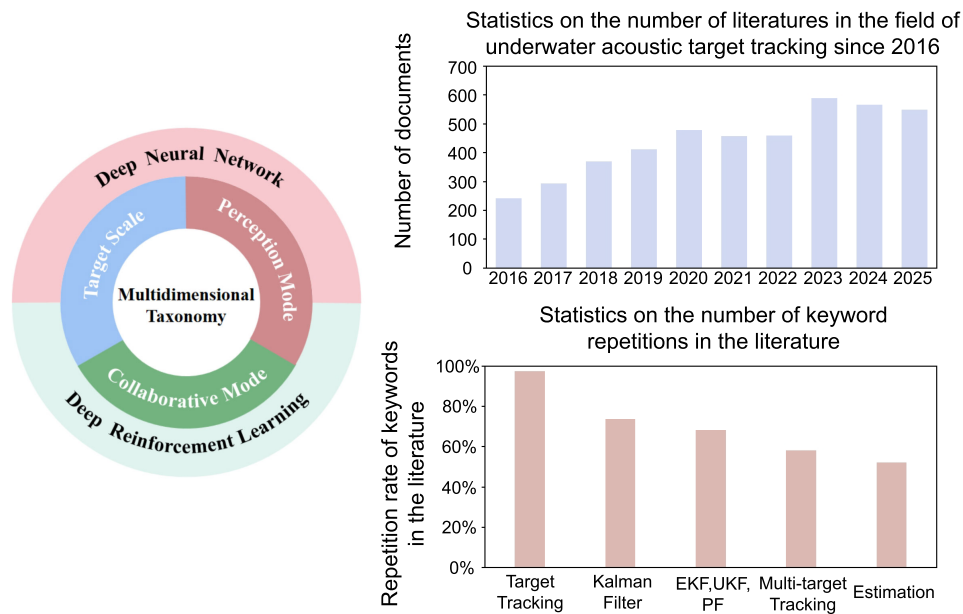


Fig. 2. Statistical analysis of the literature. We selected a comprehensive range of domestic and international literature databases, including Wanfang Database, China National Knowledge Infrastructure (CNKI), SCI, Scopus, Elsevier Science Direct, SpringerLink, Engineering Village, ACM Digital Library and IEEE Xplore. The analysis presented in this figure is based on data retrieved from these sources.

the target, perception, and processing layers of cognitive networks. Additionally, we summarize recent research progress on machine learning techniques that enhance cognitive tracking capabilities and optimize cognitive networks through deep learning, reinforcement learning, and data processing. A systematic comparison of this survey with existing reviews or commentaries is presented in Table I.

D. Contribution of the Survey

To the best of our knowledge, this survey represents the first systematic and multidimensional investigation of underwater acoustic target tracking from the perspective of cognitive communications and networking. It comprehensively reviews foundational methodologies and recent technological advancements. The primary contributions of this survey are as follows:

- **We propose a novel multi-dimensional taxonomy framework.** It integrates the three hierarchical levels of cognitive networks: the target layer, the perception layer, and the processing layer. Unlike traditional unidimensional approaches, this framework establishes a cross-mapping mechanism to elucidate the diversity and complexity of underwater acoustic tracking systems. This perspective enables a multi-faceted examination of methodologies, thereby addressing the limitations of conventional reviews.
- **We conduct an in-depth analysis of emerging cognitive technologies within tracking algorithms.** Specifically, from the perspective of cognitive communication and networks, we investigate how deep learning and reinforcement learning optimize underwater acoustic target tracking across various cognitivenetwork levels. By systematically documenting these technological advancements, this analysis addresses a significant gap in existing surveys, offering researchers both a reference

for integrating advanced technologies and a roadmap for future innovations.

- **We present a structured comparative analysis of methodological systems.** Through meticulously constructed comparison tables, technical roadmaps, and case studies, we distill the fundamental principles, distinctive features, and performance metrics of various tracking methods. This structured analytical approach not only facilitates readers' rapid comprehension of the core methodology but also provides a practical framework for technological selection and optimization in real-world applications.
- **We identify current challenges and propose future research directions.** A systematic evaluation of dynamic cognitive processes reveals critical issues related to data processing, algorithm design, and practical deployment in underwater acoustic target tracking. To address these challenges, we propose targeted research avenues, including data desensitization techniques, embodied intelligence, and model lightweighting. These directions provide a theoretical foundation for interdisciplinary research and offer actionable guidance for future studies.

This survey is structured as follows: Chapter II provides an overview of the fundamental definitions and models of cognitive underwater acoustic target tracking networks. Chapter III categorizes traditional tracking methods from three perspectives: target scale, perception method, and collaboration mode. Chapter IV explores the role of deep learning and reinforcement learning technologies in the field of cognitive underwater acoustic target tracking. Finally, Chapter V addresses current challenges and proposes future avenues and Chapter VI provides a conclusion.

As illustrated in Figure 3, this survey systematically analyzes over 170 publications spanning the period from 2016 to 2025. Utilizing the proposed multi-dimensional classification

TABLE I
THE SUMMARY AND COMPARISON OF RELATED SURVEYS

Literature	Description	Underwater acoustic Sensors				Underwater acoustic Target		Tracking methods
		Perception method: passive tracking	Perception method: active tracking	Cooperative mode: centralized	Cooperative mode: distributed	Single target	Multi target	Cognitive learning assisted
[12]	Perspective: system architecture, tracking models, energy efficiency optimization, single target and multi target tracking techniques	✗	✗	✓	✓	✓	✓	✗
[13]	Reviews concept of randomized finite set (RFS) and its application to multi-target tracking	✗	✗	✗	✗	✗	✓	✗
[14]	Perspective: instrument-assisted methods, model-based methods, and tracking optimization methods	✓	✓	✗	✗	✓	✗	✗
[15]	Reviews the application of target tracking techniques in aeronautical and underwater acoustics	✓	✓	✗	✗	✓	✗	✗
[16]	Reviews the research progress of azimuth tracking technology for underwater acoustic targets based on passive sonar systems	✓	✗	✗	✗	✓	✓	✗
Ours	New taxonomy framework: target characteristics, perception methods, collaboration modes, and cognitive learning-driven methods	✓	✓	✓	✓	✓	✓	✓

* The table describes the differences between the surveys in the field of underwater acoustic target tracking in recent years and this survey, and compares in detail the coverage of the survey contents from the perspectives of sensors, targets, and tracking methods.

framework, we categorize traditional tracking methods according to diverse task scenarios. Furthermore, we synthesize the optimizations implemented by recent machine learning-driven approaches at each layer of the cognitive network.

II. FUNDAMENTAL DEFINITIONS AND MODELS IN COGNITIVE UNDERWATER ACOUSTIC NETWORKING AND TRACKING

Underwater acoustic target tracking utilizes acoustic signals to identify and estimate the position and trajectory of a target [14]. As illustrated in Figure 1, the cognitive network architecture decomposes the tracking process into three distinct levels: the **target layer**, the **perception layer**, and the **processing layer**. This section systematically outlines the fundamental theory underlying the cognitive process of underwater acoustic target tracking across these three levels.

A. Target Layer

The target layer acts as the external driving source of the system, comprising the physical entities and their associated state sets within the cognitive network [67]. This layer primarily encompasses target motion and observation models, as well as target typology and cardinality.

1) *Target Motion and Observation Models:* Accurate underwater target tracking necessitates the establishment of

a rigorous system model [68], as mathematically formulated in Equation (1):

$$\begin{cases} x_{k+1} = f(x_k, w_k) \\ z_k = h(x_k, v_k). \end{cases} \quad (1)$$

In this model, $f(\cdot)$ represents the transition function, which is determined by the target motion model, and $h(\cdot)$ represents the measurement function, which is determined by the target observation model. x_k and z_k represent the state vector and measurement vector at time k respectively, while w_k and v_k represent the process noise vector and measurement noise vector at time k .

A widely adopted modeling approach assumes that both process noise and measurement noise are Gaussian white noise. The probability density function of the noise amplitude follows a Gaussian distribution, as defined by Equation (2):

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad (2)$$

where μ is the mean and σ^2 is the variance. In underwater acoustic channels, the noise mean is typically zero, i.e., $\mu = 0$.

In practical underwater environments, noise typically stems from multiple independent, small-scale stochastic factors. The Central Limit Theorem implies that the aggregate of numerous independent random variables converges to a normal distribution [69]. Consequently, acoustic propagation

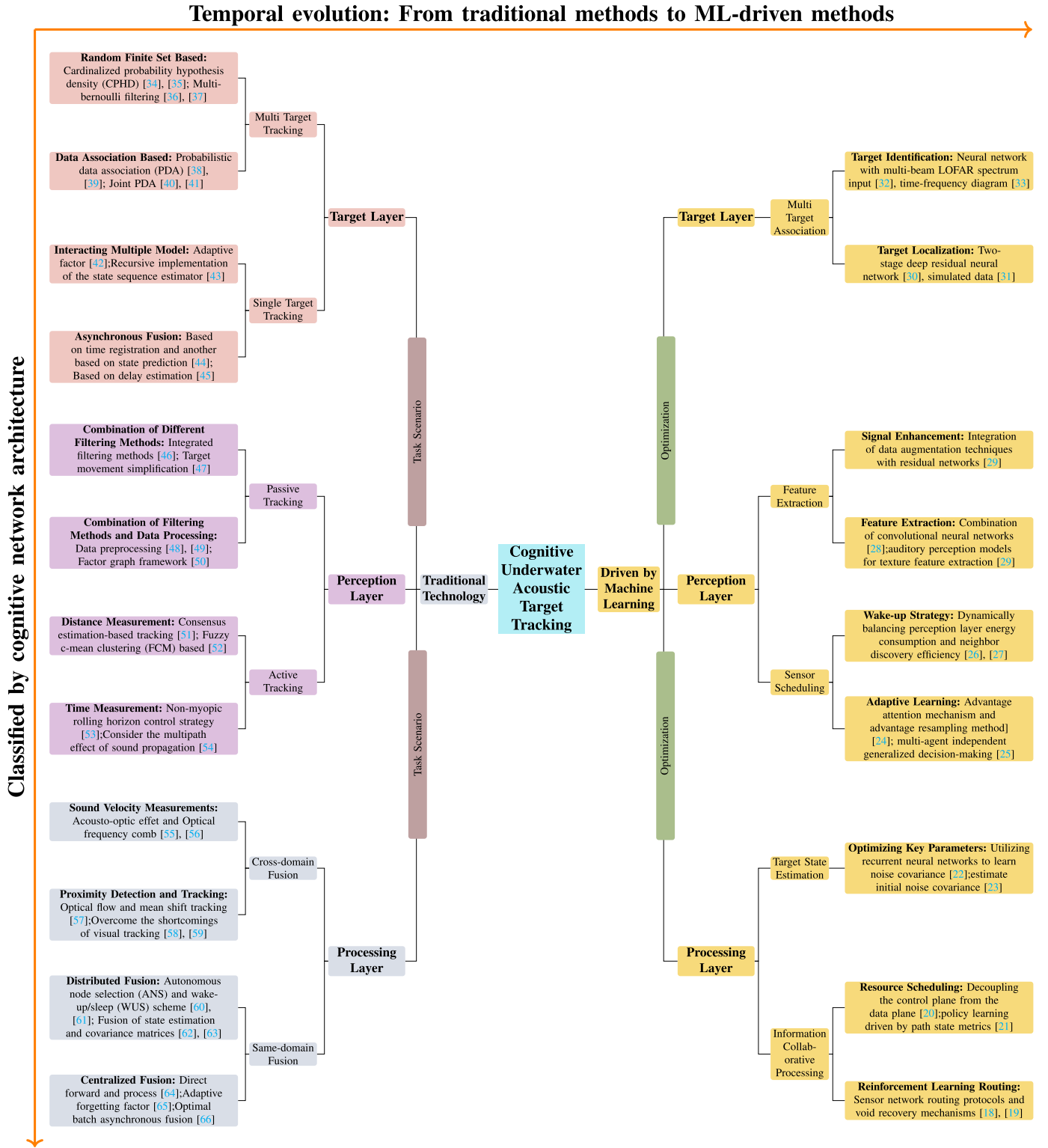


Fig. 3. Taxonomy of cognitive underwater acoustic target tracking methods. This survey synthesizes the relevant literature according to two distinct dimensions: temporal evolution and cognitive network structure. From a temporal perspective, methodologies are categorized into traditional tracking techniques and machine learning-driven approaches. In terms of cognitive network structure, a three dimensional classification framework is employed to characterize traditional tracking methods across diverse task scenarios. Furthermore, the optimizations implemented by machine learning-driven methods at various levels of the cognitive network are systematically analyzed.

noise in underwater scenarios often exhibits near-Gaussian characteristics. Even when individual noise sources deviate from a Gaussian distribution, their combined effect tends to yield an approximately Gaussian result. Moreover,

the mathematical tractability of the Gaussian distribution significantly simplifies the state transition and observation update phases in filtering algorithms. This simplification ensures computational efficiency and effectively

approximates the statistical and spectral properties of real-world noise.

The research framework for underwater target tracking algorithms fundamentally relies on two model components: (1) **target motion model**, describing the target's kinematic behavior, including constant velocity (CV) or constant acceleration (CA) model [70], time-varying models [71], semi-markov models based on Markov processes [72], current statistics models based on modified rayleigh-markov processes [73], and jitter models incorporating acceleration derivatives [74]; and (2) **target observation model**, characterizing the measurement process, including time-of-arrival (TOA)-based, received signal strength (RSS)-based, and angle-of-arrival (AOA)-based measurement models [75], [76].

2) *Multi Target Association*: Modeling target typology and cardinality provides essential prior information for the cognitive network system. Distinct target classes—such as submarines, autonomous underwater vehicles (AUVs), schools of fish, or operational equipment—exhibit unique acoustic scattering properties, maneuverability profiles, and spectral characteristics. These variations dictate the selection of appropriate perception and tracking algorithms.

In single-target scenarios, high-precision state estimation is achievable using continuous observation sequences. Conversely, multi-target environments present challenges such as signal superposition, mutual interference, and occlusion. Addressing these complexities requires multi-target modeling and data association mechanisms capable of characterizing both the independence and coupling among targets. Furthermore, as target cardinality varies over time, the system must dynamically detect the appearance of new targets and the disappearance of existing ones to maintain the robustness and adaptability of the cognitive network.

B. Perception Layer

The perception layer facilitates environmental observation, signal acquisition, and feature extraction within the cognitive network system. By capturing signals emitted by underwater targets, it directly determines the system's detection performance and signal quality [77]. Key components of this layer include underwater acoustic signal propagation, signal acquisition and preprocessing, and sensor cluster deployment.

1) *Propagation of Underwater Acoustic Signals*: Underwater acoustic signal propagation can be characterized using either Received Signal Strength (RSS) [78] or wave equation modeling. Without accounting for the complexity of the marine environment, RSS magnitude can be expressed as a function of the straight-line distance between the transmitter and receiver [79]. Recent studies have increasingly emphasized the marine environment as a critical factor in underwater acoustic signal propagation. To accurately characterize sound propagation in water, researchers now strive to comprehensively account for the propagation medium.

The majority of fundamental theories addressing underwater acoustic propagation problems—including **ray theory** [80], [81], **normal mode theory** [82], and **parabolic equation theory** [83]—are derived from wave fluctuation equations. These theoretical frameworks provide distinct approaches to

modeling sound propagation in underwater environments, each with specific advantages and limitations.

2) *Signal Acquisition and Preprocessing*: Signal acquisition and processing within the perception layer necessitate multi-node synchronous sampling to ensure temporal and amplitude consistency. During the preprocessing stage, techniques such as denoising and adaptive beamforming are employed to suppress background noise and interference, for enhancing the signal-to-noise ratio (SNR). Furthermore, the cognitive system executes multi-dimensional feature extraction on the acquired signals [84]. These features not only facilitate target classification and identification but serve as high-quality inputs for state estimation and decision optimization in the layer.

3) *Deployment of Sensor Clusters*: To achieve efficient and accurate perception, the deployment of sensor clusters is critical. The perception layer leverages tailored array structures and node distribution strategies to enable spatial sampling and high directional resolution. For instance, linear, planar, and spherical arrays offer distinct spatial scanning capabilities, while distributed underwater acoustic sensor networks provide wide-area coverage and robustness through collaborative node detection. To enhance detection reliability, array configurations and inter-element spacing must be optimized based on operating frequency, target range, and desired angular resolution.

C. Processing Layer

The processing layer serves as the intelligent core of the system, responsible for intensive computation, state estimation, and global collaboration. This layer enables closed-loop processing that transforms perceptual data into decision-making results [85]. It primarily includes target state estimation methods and collaborative mechanisms within the cognitive network.

1) *State Estimation of Underwater Acoustic Targets*: State estimation methods within underwater acoustic target tracking are usually classified into two categories: batch methods and recursive methods. The fundamental differences in principle between these approaches are illustrated in Figure 4.

1-1) Batch Processing Method

Batch processing methods process all observations collected during a specific time interval as a single dataset to estimate the target state. Representative methods include maximum likelihood estimation (MLE) [86], least squares estimation (LS) [87], gauss-newton method (GN) [88], pseudo linear estimation (PLE) [89], and monte carlo method (MC) [90].

MLE determines the optimal target state by maximizing the likelihood function, which quantifies the probability of observed data given state parameters [91]. LS estimation minimizes the sum of squared residuals between observations and model predictions through optimization [92]. The GN method iteratively solves nonlinear least squares problems by linearizing the model at each step [93]. PLE simplifies nonlinear problems via linear transformations or auxiliary variables [94], while MC uses random sampling to approximate posterior distributions and quantify uncertainty [95].

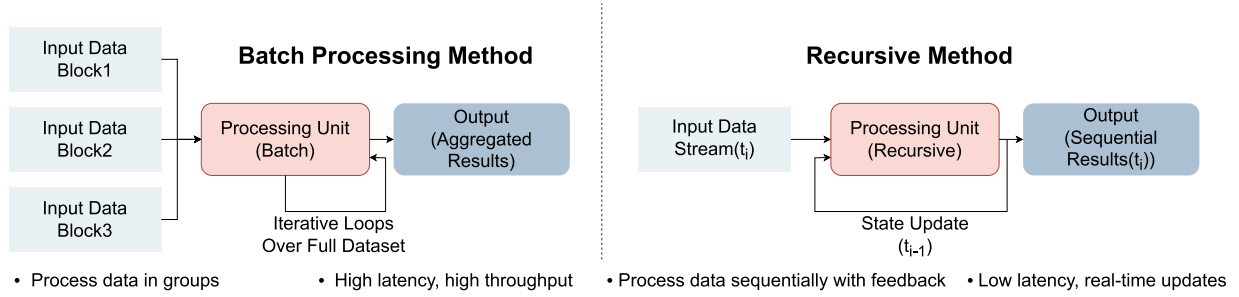


Fig. 4. Schematic illustration of the workflows for batch processing and recursive methods. Batch processing methods process data in groups, whereas recursive methods focus on the relationship between the previous and current time steps.

1-2) Recursive Method

However, batch methods exhibit two main limitations. First, they require complete data collection before processing, precluding real-time state updates. Second, their computational efficiency decreases significantly with large datasets. These limitations become particularly problematic given recent advancements in monitoring technologies and improvements in target maneuverability, which have increased tracking uncertainties [14].

A primary challenge in designing target tracking algorithms to address the specific constraints of underwater environments, governed by established dynamic motion and observation models. Most filtering methodologies rely on recursive estimation theory, with Bayesian filtering serving as the prominent paradigm.

The fundamental principle of the Bayesian filter is the utilization of current observational data to correct the prior probability density function (PDF) [96], thereby yielding the posterior probability. This framework comprises two distinct stages: prediction and update. The prediction stage derives the prior PDF based on the target motion model, while the update stage incorporates measurements from the observation model to refine the prediction, resulting in the posterior probability of the target state.

The Kalman Filter (KF) provides an optimal Bayesian solution under linear Gaussian conditions [97]. However, in nonlinear systems, deriving an optimal solution becomes intractable due to the computational complexity associated with infinite-dimensional integrals. Systems where nonlinear dynamics can be locally linearized with negligible higher-order Taylor expansion terms are characterized as weakly nonlinear, whereas those with significant higher-order terms exhibit strong nonlinearity.

To address these nonlinearities, the Extended Kalman Filter (EKF) employs a first-order Taylor approximation for state estimation [98]. Alternatively, the Unscented Kalman Filter (UKF) preserves higher-order statistical information via sigma-point transformation [99]. For strongly nonlinear systems, the Particle Filter (PF) utilizes Monte Carlo sampling to approximate state probability distributions, often demonstrating superior performance [100].

2) *Cognitive Network Collaboration*: To enhance the efficiency of data analysis and processing, underwater acoustic target tracking has evolved from single-sensor

operations to multi-sensor collaborative frameworks. Distributed sensor nodes synchronize status information and exchange feature data via underwater acoustic communication links. Given the inherent challenges of underwater acoustic channels—specifically high latency, limited bandwidth, and severe multipath effects—cognitive systems must employ efficient signal transmission mechanisms. Techniques such as time-frequency division multiplexing, coherent Orthogonal Frequency Division Multiplexing (OFDM), and spread spectrum communication are essential to ensure reliable data transmission [101].

The cognitive processing layer extends beyond data fusion and state estimation to include adaptive optimization and closed-loop feedback capabilities. By analyzing tracking accuracy, channel conditions, and node energy consumption in real time, the system optimizes the allocation of communication power, computational resources, and tasks [102]. This dynamic collaborative control enables the cognitive network system to sustain stable and efficient tracking performance, even in complex and time-varying marine environments.

III. CLASSIFICATION OF TRADITIONAL UNDERWATER ACOUSTIC TARGET TRACKING METHODS

Underwater acoustic target tracking represents a multifaceted and dynamic research domain wherein methodologies are typically distinguished by diverse criteria. A systematic taxonomy and synthesis of these approaches establish a critical foundation for in-depth investigation. In this chapter, methods are categorized according to the three tiers of the cognitive network: the **target layer**, **perception layer**, and **processing layer**. Specifically, the classification framework encompasses **target scale**, **perception method**, and **collaboration mode**.

A. Target Tracking Methods Distinguished by Target Scale

Underwater acoustic target tracking involves analyzing two key target characteristics: physical dimensions and sensor interaction capabilities. In most practical scenarios, targets exhibit non-cooperative behavior by avoiding active sensor engagement. This section classifies tracking methodologies according to target quantity, subsequently detailing both **single target** and **multi target** tracking approaches.

1) *Single Target Tracking*: It plays a pivotal role in underwater acoustics, where research is primarily directed toward

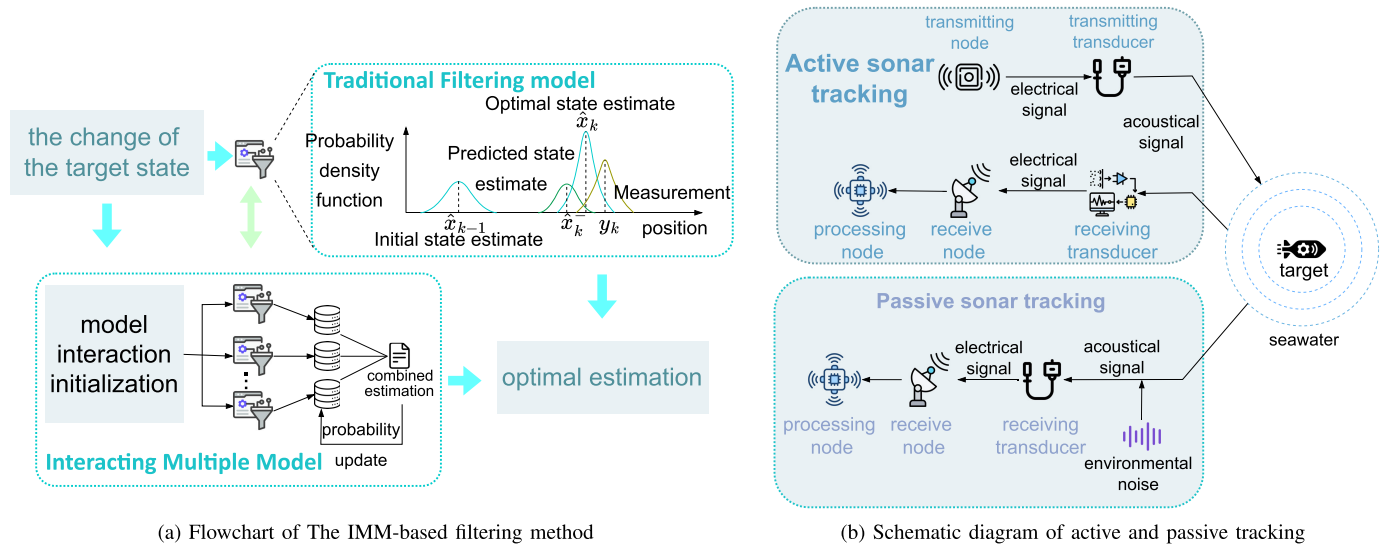


Fig. 5. Illustration of the IMM tracking paradigm and tracking modes. (a) The paradigm employs multiple base filtering models to estimate the target state, combining their estimation results with specific weights. These weights are then adjusted based on the final estimation outcome for each filtering model. (b) In active tracking, acoustic signals are actively emitted, and the tracking process begins at the transmitting node and ends at the receiving node. In contrast, passive tracking does not actively emit acoustic signals; instead, it relies solely on the receiving node to capture acoustic signals from the external environment.

enhancing tracking accuracy and response speed for dynamic targets [103]. The **Interactive Multiple Model (IMM)** algorithm is utilized to address model mismatches arising from the variability of target motion states, while **asynchronous fusion** techniques are employed to mitigate data asynchrony caused by propagation delays and environmental interference.

This section comprehensively reviews the current state and technological advancements in single target tracking, with a specific emphasis on the application of IMM algorithms and the critical role of asynchronous fusion in optimizing tracking performance.

1-1) IMM algorithm

The IMM algorithm employs multiple models to represent potential target motion states, with Markov processes governing transitions between these models. During tracking, the algorithm establishes parallel model filters corresponding to possible target states for real-time maneuvering detection. By assigning weight coefficients and model update probabilities to each filter, the system generates optimal state estimates through weighted computations, thereby achieving model-adaptive tracking [104]. Figure 5a illustrates the operational flowchart of this algorithm.

A variable-structure IMM was proposed in [105], establishing theoretical foundations for optimal estimators. Chen et al. [42] introduced adaptive factors enabling real-time model state switching based on acceleration and velocity data, simultaneously improving tracking accuracy while reducing computational load. The recursive IMM algorithm [43] implemented a state sequence estimator recursively, simplifying computations by incorporating Kalman filter outputs in weight calculations. A parameter-adaptive IMM [106] combines parameter identification to enable real-time estimation of filtering parameters including mode transition probabilities and model noise variance. Tan et al. [106] replaced conventional probability calculations with fuzzy logic reasoning, proposing

a Fuzzy-logic based maneuvering target tracking algorithm that enhances model matching accuracy through normalization processing, demonstrating significant practical utility.

1-2) Asynchronous fusion techniques

The acoustic signal's slow propagation speed and high susceptibility to marine environmental interference, combined with communication latency between sensors, make it challenging to achieve fully synchronized target measurements. This asynchrony manifests in two distinct fusion scenarios: in-sequence measurement (ISM) and out-of-sequence measurement (OOSM) fusion [44]. ISM occurs when the fusion center receives measurements in temporal alignment with the target's acoustic signal emission sequence, whereas OOSM arises when these sequences are misaligned.

Despite existing asynchronous fusion localization algorithms for underwater sensors [107], [108], research on asynchronous fusion tracking remains limited. Current solutions for measurement asynchrony caused by sampling frequency disparities primarily employ time registration techniques to synchronize data. For instance, Yang et al. [44] compared time-registration-based and state-prediction-based fusion algorithms, while Liu et al. [45] introduced a delay-estimation asynchronous particle filter that improves weight updating through backpropagation. Furthermore, Yan et al. [109] presented an innovative approach that establishes propagation delay-position relationships to compensate for clock offsets, subsequently developing a consensus Bayesian filter for continuous target tracking. This solution enhances both tracking accuracy and network longevity through combined consensus fusion and duty-cycle mechanisms.

2) *Multi Target Tracking*: With the advancement of research in underwater acoustic target tracking, multi-target tracking has emerged as a critical research focus. Considering current marine environment, future targets are expected to predominantly appear in clustered formations with enhanced

TABLE II
COMPARISON OF VARIOUS DATA ASSOCIATION BASED METHODS

Method		JML [111]	TSM [112]	GNN [113]	PDA [38], [119], [120]	MHT [114], [115], [118]	JPDA [40], [41]	JIPDA [41], [121]
Advantages	Theoretical optimization	✓	/	/	/	/	/	/
	Low computational complexity	/	/	✓	/	/	/	/
	Track management	/	✓	/	/	✓	/	✓
	Robustness in the presence of dense targets	/	/	/	✓	✓	✓	✓
	Adaptation when target are added	/	/	/	/	/	/	✓
Disadvantages	High computational complexity	○	/	/	/	○	○	○
	High storage requirements	/	○	/	/	○	/	/
	Weak anti-interference capability	/	/	○	○	/	/	/
	Poor robustness when targets are dense	/	/	○	○	/	○	/
Applicable Scenarios		Low noise with well-aligned observations and trajectories	Temporarily intersect trajectories or high uncertainty	Few sparsely distributed targets	Non-intersecting trajectories with low clutter ratio	Dense targets with highly intersecting trajectories	Dense targets under moderate noise conditions	Well-defined parametric models with limited targets

* The table summarizes the advantages, disadvantages, and applicability scenarios of the seven data association based approaches, where "✓" denotes the method has this advantage ; "○" denotes the method has this disadvantage; "/" indicates the absence of the corresponding advantage or disadvantage.

maneuverability and improved concealment capabilities. Furthermore, the complex marine environment and fluctuating underwater acoustic channels often generate substantial clutter, while measurements may be contaminated by uncertainties including false alarms and ambient noise [110].

In these challenging underwater conditions, accurately estimating both the number of targets and their respective states based on sensor node measurements remains a fundamental challenge for underwater tracking systems. Consequently, multi-target tracking technology has become a pivotal research direction in this field. Presently, multi-target tracking methodologies can be broadly categorized into two approaches: **data association-based techniques and random finite set (RFS)-based methods**.

2-1) Data association-based tracking

This approach employs data association algorithms to establish correspondences between targets and measurements. By integrating Bayesian filtering frameworks, the complex multi-target tracking problem is decomposed into manageable single-target tracking subproblems. The process fundamentally comprises two sequential stages: (1) data association and (2) state filtering [12].

Since state estimation validity depends on accurate data association, these techniques constitute the cornerstone of traditional multi-target tracking systems. Current methodologies are broadly categorized into:

Maximum Likelihood Methods: joint maximum likelihood (JML) [111] and track splitting method (TSM) [112], which optimize constructed likelihood functions through batch processing.

Bayesian Methods: global nearest neighbor (GNN) [113], multiple hypothesis tracking (MHT) [114], [115], probabilistic MHT (PMHT) [116], [117], [118], probabilistic data association (PDA) [38], [119], [120], and their derivatives including JPDA [40], integrated JPDA [121], comprehensive JPDA [122], trajectory information-based JPDA [41] and TJPDA [41]. These recursive approaches enable real-time state estimation.

In the complex underwater acoustic environment with challenging signal processing conditions, **Bayesian methods have become predominant for multi-target tracking in underwater acoustic applications**. Table II systematically compares the trade-offs and applicability of these association techniques.

2-2) RFS-based tracking

The RFS-based multi-target tracking algorithm models the multi-target density as an RFS. Within the Bayesian framework, the FISST-based Bayesian filtering algorithm iteratively propagates the RFS density, effectively transforming the multi-target tracking problem into a mapping problem between random sets of target states and measurements. This approach simultaneously estimates both the number of targets and their states [123].

However, the combinatorial nature of multi-target probability density and the multiple integrals in high-dimensional multi target state spaces make the implementation of an optimal RFS-based multi target Bayesian filter highly challenging. **Consequently, approximate filtering methods, such as probability hypothesis density (PHD) filter and multi-bernoulli (MB) filter, have been developed as principled approximations.**

TABLE III
COMPARISON OF VARIOUS RFS BASED METHODS

Method		PHD [125]–[127]	CPHD [34], [35], [128]	MB [124]	LMB [36], [129]	PMBM [130], [131]
Advantages	Low computational complexity	✓	/	✓	/	/
	Quantity estimates	/	✓	/	/	/
	track management	/	/	/	✓	✓
	High computational efficiency	/	/	/	/	✓
	Adaptation when target are added	/	/	/	/	✓
Disadvantages	High computational complexity	/	○	/	○	○
	Poor robustness when targets are dense	○	/	○	/	/
	Difficulty in dealing with nascent targets	/	○	/	/	/
Applicable Scenarios		Scenarios with dynamic target count and low computational resources	Scenarios requiring stable target count and precise estimation	Scenarios with sparse targets requiring identity distinction	High-precision scenarios with dense targets and crossing trajectories	Scenarios with clutter and newly emerging targets

* The table summarizes the advantages, disadvantages and applicability scenarios of the five RFS based approaches, where "✓" denotes the method has this advantage; "○" denotes the method has this disadvantage; "/" indicates the absence of the corresponding advantage or disadvantage.

The PHD filter assumes that the multi-target probability density follows a Poisson RFS distribution, defined as:

$$\pi_{k|k}(X) = e^{-\lambda} \prod_{x \in X} v_{k|k}(\mathbf{x}), \quad (3)$$

where λ represents the mean number of targets in the multi-target state X , and the target number distribution $\rho(n)$ follows a Poisson distribution with parameter λ .

The MB filter [124] employs a parametric approximation method. Specifically, at each time step k , the multi-target state can be mathematically represented as a union of multiple independent Bernoulli RFS, as formalized in Equation (4):

$$X = \bigcup_{i=1}^{n_k} X_i. \quad (4)$$

The density function of X_i is represented by the Bernoulli RFS distribution:

$$\pi_{k|k}(X_i) = \begin{cases} 1 - r_{k|k,i}, & \text{if } X_i = \emptyset \\ r_{k|k,i} \cdot p_{k|k,i}(\mathbf{x}), & \text{if } X_i = \{\mathbf{x}\} \\ 0, & \text{else,} \end{cases} \quad (5)$$

where $r_{k|k,i} \in (0, 1)$ represents the probability of existence for the i -th target. According to the definition above, the multi-target probability density can be completely represented by the MB parameter set $\{r_{k|k,i}, p_{k|k,i}(\mathbf{x})\}_{i=1}^{n_k}$. The standard MB filtering algorithm recursively propagates the MB parameter set to replace the full multi-target probability density, thereby approximating the multi-target Bayesian filtering process.

Table III compares the advantages, disadvantages, and limitations of various multi-target tracking methods based on RFS.

B. Target Tracking Methods Distinguished by Perception Method

Sensor perception modalities are primarily classified into active and passive perception. Active perception involves sensors emitting detection signals and receiving the corresponding echoes, whereas passive perception entails solely receiving signals emitted by targets for non-contact sensing. Based on these two perception modalities, tracking algorithms can be categorized into **active tracking** algorithms and **passive tracking** algorithms.

Active tracking algorithms aim to estimate the target state by utilizing sensors such as active sonar, which provide information on the azimuth and range of the target based on the echoes received. These echoes originate from the system itself [132]. Passive tracking utilizes sensors such as passive sonar to receive and analyze signals radiated by the target in order to track it, without transmitting detection signals. This approach avoids issues such as exposing the tracker's information and the energy consumption associated with signal transmission [133]. The schematic of active and passive sonar perception is shown in Figure 5b.

1) *Active Tracking*: Active sonar systems play a pivotal role in underwater target detection and tracking, comprising three essential components: a transmitter, a receiver, and a signal processing unit. The transmitter periodically emits designed acoustic pulses, while the receiver captures the corresponding echoes. Subsequently, the processing unit analyzes these echoes to extract critical target parameters including position, velocity, and range [134]. Nevertheless, active tracking presents significant operational risks as the

TABLE IV
COMPARISON OF THE CHARACTERISTICS AND SPECIFIC METHODS OF ACTIVE AND PASSIVE TRACKING

Mode	Advantages	Disadvantages	Characteristics	Literature	Tracking Method
Active tracking	High precision, controllable signal frequency, long-range detection	Prone to location exposure, high energy consumption	Proactive path planning for observation	[52]	Control strategy for AUV a branch and bound technique
			Suitable for maneuvering targets	[54]	FCM,KF
			Optimizing network energy efficiency	[51] [51]	EKF,UKF CEUTT
Passive tracking	Strong concealment, low energy consumption	Low precision, susceptible to environmental interference	Handling heavy noise	[49]	Pre-processing of noise, UKF, IUKF
				[137]	Sage-Husa,EKF
			Multi-sensor scheduling	[46]	EKF,factor graph
				[138]	Sage-Husa,APF
				[49]	EKF,UKF
				[139]	VB,KF
			Integrating auxiliary information	[140]	Sage-Husa,EKF
				[46]	consensus estimation, IUKF
				[141]	EF,EKF

* The table describes the advantages and disadvantages of active and passive tracking, along with a list of relevant literature and a summary of the tracking methods used in the literature.

emitted signals may reveal the sensor's location, particularly in complex combat scenarios, potentially compromising system security [135]. This inherent vulnerability has resulted in limited research attention in this domain.

While [52] and [53] focused on the development of a bistatic sonar system, [51] and [54] expanded the sensor configuration, integrating localization information from multiple sensors through advanced data fusion mechanisms.

2) *Passive Tracking*: Passive tracking systems exhibit three notable advantages: extended detection range, enhanced operational safety, and reduced energy consumption. Nevertheless, underwater target tracking faces considerable challenges due to dynamic environmental variables, particularly water depth, salinity variations, and ocean currents [136].

Recent advancements in conventional tracking methods, such as the KF, have directed research toward two primary areas: (1) **enhancing tracking accuracy in noisy marine environments by integrating filtering algorithms with advanced data processing techniques** [48], [49], [50]; and (2) **optimizing multi-target tracking performance by combining diverse filtering approaches** [46], [47], [49]. These developments contribute significantly to the field, offering both theoretical foundations and practical guidelines for achieving efficient and secure underwater monitoring.

Table IV compares the two perceptual modes and also lists the corresponding articles in order to give the reader a quick overview of the characteristics of the different perceptual modes and the general application scenarios of each paper.

C. Target Tracking Methods Distinguished by Collaborative Mode

When utilizing sensors for hydroacoustic target tracking, a single sensor can process the signals emitted by the target and estimate the target state, but the tracking accuracy is limited due to the very limited energy resources, bandwidth resources, and data processing capabilities of a single sensor.

Therefore, it is considered to utilize the collaboration between multiple sensors to improve the tracking performance of the system and make the system more robust and reliable. The collaboration modes of sensors can be classified into **same-domain fusion** and **cross-domain multimodal fusion** based on the type of sensors. Table V provides a detailed comparison of the characteristics of these three collaborative modes.

1) *Same-Domain Fusion*: Same-domain fusion encompasses two primary approaches: (1) **Centralized fusion**, which involves transmitting measurement data from individual sensors to a central processing unit (fusion node) for unified processing and analysis; and (2) **Distributed fusion**, where multiple autonomous sensors or nodes share and process information while maintaining independent computing and decision-making capabilities.

1-1) Centralized fusion

This collaborative approach maximizes the utilization of signal source information, thereby minimizing system information loss and achieving relatively high tracking performance. However, this architecture demands substantial communication bandwidth and exhibits reduced system reliability [12].

Previous studies have employed various centralized fusion approaches for underwater target tracking. In [64], sensor nodes directly transmit collected measurement data to a fusion center, where algorithms such as the EKF are utilized for high-precision state estimation through measurement model correction and node depth adjustment mechanisms. A similar centralized processing architecture is adopted in [103], where all sensor measurements are aggregated at the fusion node and processed using Gaussian orthogonal basis linearization of nonlinear functions to achieve accurate state estimation.

Qiu et al. [65] introduced two significant improvements to the conventional Centralized Fusion KF: (1) an adaptive forgetting factor was incorporated to enhance filter performance, and (2) an IMM approach was integrated with the

TABLE V
COMPARISON OF THE FEATURES OF THREE COLLABORATIVE MODES

Collaborative mode	Centralized Fusion	Distributed Fusion	Cross-domain Multimodal Fusion
Literature	[12] [64] [103] [65] [142] [66]	[12] [143] [60] [61] [62] [63] [135]	[12] [144] [57] [58] [59] [145] [55] [56]
Data Processing	All data converges to fusion node for centralized processing	Nodes independently process & partially fuse data, share final results	Fuses multi-type sensor data, requires solving spatiotemporal alignment & feature fusion challenges
Communication Needs	High: Full raw data transmission to center	Low: Only local estimates & essential info exchange	Medium: Requires synchronized multimodal streams & protocol/format conversion
Scalability	Low: Computational/communication load increases with sensor expansion	High: Easy node addition with strong local processing	Medium: New sensor types require algorithm adaptation & hardware compatibility
Fault Tolerance	Low: System failure upon central node malfunction	High: Single node failure has minimal system impact	Medium: Relies on redundant modalities with modality missing robustness algorithms
Real-time Performance	Low: Centralized processing induces latency	High: Local decision-making with low fusion delay	Medium: Additional computation time for multimodal fusion (hardware-dependent)
Computational Load	High: Demands high-performance computing at central node	Low: Distributed computation across nodes	High: Requires complex algorithms on fusion nodes/edge devices
Management	Centralized control for unified optimization	Decentralized coordination with synchronization complexity	Hybrid management requiring multimodal sensor parameter synchronization
Data Consistency	Easily maintains global consistency	Requires feedback mechanisms to calibrate local estimation deviations	Demands strict timestamp synchronization & coordinate alignment across modalities

* The table compares the characteristics of centralized fusion, distributed fusion, and cross-domain multimodal fusion and gives the rank, and also lists the literature corresponding to the three fusion approaches.

adaptive KF. Addressing the challenge of asynchronous measurements, Hu et al. [142] developed a novel fusion algorithm suitable for general asynchronous multi-rate sensor systems. Notably, their proposed algorithm imposes no constraints on sensor quantity, sampling rates, or initial sampling times.

While centralized fusion architectures theoretically offer comprehensive data utilization by transmitting all measurements to a fusion node, practical implementation in underwater acoustic target tracking faces significant limitations. These include constrained communication bandwidth, limited sensor endurance, and restricted computational resources, which collectively result in reduced tracking accuracy and poor engineering feasibility.

1-2) Distributed Fusion

Distributed fusion systems are distinguished by their feedback mechanisms and operational efficiency [146]. As illustrated in [143], researchers developed a feedback-based underwater distributed fusion algorithm that integrates radar system feedback mechanisms, allowing local sensors to receive information from the fusion center. Similarly, Yu et al. [60] proposed a distributed target tracking algorithm utilizing an IMM filter with a wake-up/sleep scheme, which selectively activates only those sensors within the target's activity area. These activated sensors conduct local state estimation and subsequently generate globally optimal estimates through the IMM method.

The core of distributed fusion lies in integrating node data to form local estimates. For instance, Hare et al. [61] described how sensor nodes utilize a collaborative decision processing

algorithm to fuse target state estimates, covariance matrices, and classification information from neighboring nodes, thereby producing local fusion results. Further developments include [62], [135], where sensor nodes receive broadcast signals via acoustic modems and employ track-to-track fusion to integrate state estimates and covariance matrices from multiple sensors.

2) *Cross-domain Multimodal Fusion:* Acoustic sensors demonstrate excellent long-range detection capabilities with strong penetration performance even in turbid water. Nevertheless, they suffer from relatively low resolution and are susceptible to interference from marine biological noise and multipath effects. In contrast, optical sensors can capture high-resolution images at close ranges, allowing for detailed target feature extraction [144]. However, their performance is heavily dependent on lighting conditions and deteriorates significantly in turbid water or low-light environments. Consequently, integrating optical and acoustic sensors for underwater target tracking can effectively leverage the complementary advantages of both modalities.

This section systematically examines **optical tracking methodologies for underwater targets**, followed by an analysis of **current applications involving acousto-optic sensing mechanisms in underwater target tracking**. Specifically, the discussion focuses on seawater sound velocity measurement techniques employing acousto-optic sensing principles.

In the field of close-range underwater target tracking, optical sensors have been extensively employed, predominantly utilizing visual analysis and underwater imaging techniques. As demonstrated in [57], researchers have evaluated the

applicability of two prevalent visual tracking algorithms—optical flow tracking and mean shift tracking—in underwater environments.

Subsequent studies have focused on addressing these limitations [58], [59]. Chuang et al. [58] introduced a novel multi-fish tracking algorithm incorporating feature-based temporal matching and Viterbi data association. This approach effectively resolves challenges posed by rapid fish movements and frequent occlusions in low-frame-rate conditions.

To address nighttime tracking challenges, Snyder et al. [145] proposed an innovative imaging technique as an alternative to acoustic cameras for underwater tracking. This method employs electrical impedance tomography with self-generated electric fields.

The exceptional measurement accuracy and non-contact nature of acousto-optic sensing offer substantial potential for oceanographic applications. Integrating acousto-optic methods into underwater target tracking presents not only technical challenges but also interdisciplinary system engineering opportunities, opening new possibilities for deep-sea exploration. Continued progress in data fusion technologies, sensor development, and computational power will undoubtedly enhance and promote wider adoption of acousto-optic hybrid underwater tracking systems.

D. Summary

This chapter provides a three-layered and three-dimensional methodological system from three different perspectives: target scale, perception approach, and collaboration mode. From the perspective of target scale, tracking methods can be divided into single-target tracking and multi-target tracking. From the perspective of perception approach, tracking methods can be divided into active tracking and passive tracking. From the perspective of collaboration mode, tracking methods encompass centralized, distributed, and cross-domain multimodal fusion approaches.

IV. EMERGING TRENDS IN TRACKING METHODS DRIVEN BY MACHINE LEARNING

Chapter III established the fundamental role of traditional tracking methodologies within the three-layer architecture of cognitive underwater acoustic target tracking networks. Building on that foundation, this chapter shifts focus to machine learning (ML) technologies. Given the inherent limitations of traditional approaches in complex underwater environments, ML techniques—particularly Deep Neural Networks (DNN) and Deep Reinforcement Learning (DRL)—are advancing the field toward a higher level of cognitive capability [147]. Distinguished by their robust nonlinear modeling and adaptive learning mechanisms, these technologies enable data-driven feature extraction and policy optimization. Consequently, they effectively unveil latent patterns within complex acoustic signals [148], facilitating critical breakthroughs across the three core dimensions of traditional tracking methods.

Specifically, this chapter analyzes how machine learning addresses three major bottlenecks encountered by conventional approaches:

- **Target Scale:** Traditional algorithms, such as KF, PDA, and MHT, often struggle with nonlinear modeling and the combinatorial explosion of data association in multi-target scenarios. To mitigate these issues, we introduce the high-dimensional feature extraction capabilities of DNNs and the intelligent decision-making frameworks of DRL to enhance tracking robustness.
- **Perception method:** Conventional active and passive sensing methods are constrained by fixed resource allocation and limited feature extraction capabilities. To overcome these drawbacks, we leverage the active control mechanisms of DRL to realize “cognitive sensing,” while utilizing DNNs to strengthen signal processing depth.
- **Collaboration Mode:** Co-located and cross-domain fusion strategies frequently face challenges regarding heterogeneous data alignment and high communication overheads. We address these challenges by adopting DNN-based multimodal fusion architectures and DRL-driven distributed scheduling strategies to achieve efficient collaboration.

Based on this optimization logic, this chapter explores the theoretical advancements and practical applications of DNN and DRL in cognitive underwater acoustic target tracking. It provides a comprehensive overview of novel tracking methods driven by these technologies, highlighting their specific impact on optimizing the cognitive network’s three-layer structure. Furthermore, the chapter synthesizes the paradigm shifts that machine learning brings to traditional underwater acoustic target tracking frameworks.

A. Target Layer

1) Deep Neural Networks for Capturing Multi Target Correlated Information: DNN technologies are capable of processing complex acoustic signals to identify key correlated information [149]. Consequently, DNNs are widely applied in underwater target recognition and localization [150].

In multi-target environments, these techniques generate accurate initial state labels, thereby enhancing the efficiency of data association within subsequent perception and processing layers. Additionally, DNNs address target loss issues inherent in traditional tracking methods, improving overall tracking robustness.

1-1) Target Localization

Niu et al. [30] and Huang et al. [31] developed a two-stage model based on simulated data. In this approach, feature vectors derived from the modal signal subspace decomposition of covariance matrices serve as inputs to a DNN for range and depth regression. To further advance the field, Huang et al. [31] implemented a multi-task learning framework featuring adaptive weighted loss within a Convolutional Neural Network (CNN) architecture to enhance deep-sea acoustic source localization.

1-2) Target Identification

Wang et al. [32] integrated multi-beam Low-Frequency Analysis and Recording (LOFAR) data [151] with an improved LeNet-5 CNN, combined with an EKF, to further improve

tracking accuracy. Similarly, Wang et al. [33] incorporated target azimuth information into the time-frequency spectrogram to mitigate target loss in traditional tracking systems.

2) *Comparison and Analysis of Performance Metrics:* DNN-based tracking methods for target layer optimization encompass various scenarios, including shallow-water versus deep-water, active versus passive, and single-sensor versus array configurations. To compare these methods, key performance indicators were selected according to the core task types:

2-1) *Mean Absolute Error (MAE)*

Shallow-water approaches [30], [31] exhibit higher accuracy, achieving a range MAE of less than 0.2 km. This performance is primarily due to the simpler sound propagation paths and minimal environmental parameter fluctuations characteristic of shallow water. Conversely, deep-water approaches [152] achieve slightly lower accuracy, with a range MAE of 1.3 km, but demonstrate greater adaptability. The reduced accuracy is attributed to multi-regional superposition effects in deep-water acoustic propagation. However, these deep-water methods effectively address issues such as array tilt and depth mismatch through multi-task learning, thereby ensuring coverage of complex deep-water acoustic field regions.

2-2) *Bias*

Active tracking methods [33] demonstrate smaller bias (≤ 200 m). Because active sonar emits signals, its combination with Deep Convolutional Neural Network (DCNN) identification of time-frequency spectrograms enables stable tracking even under low SNR conditions. In contrast, passive tracking methods [32] yield lower azimuth error (1.066°). Leveraging the strong concealment of passive sonar, these methods enhance azimuth features using multi-beam LOFAR spectra, which helps avoid track loss caused by azimuth overlap among multiple targets.

B. Perception Layer

1) *Deep Neural Networks for Perception Optimization:*

The application of DNNs within the perception layer primarily focuses on automatic feature extraction from complex underwater acoustic signals and signal enhancement against high-noise backgrounds [153]. While traditional beamforming and filtering methods rely on manually designed algorithms, end-to-end acoustic feature extraction approaches based on DNNs—such as Convolutional Neural Networks (CNNs) or Transformers—can learn highly discriminative features directly from raw acoustic waveforms. This capability eliminates the need for labor-intensive manual feature engineering. Additionally, DL-driven denoising networks effectively suppress environmental noise and multipath interference under low SNR conditions, thereby providing the processing layer with high-fidelity input data [154].

Yang et al. [28] combined CNNs with auditory perception models, utilizing Mel Frequency Cepstral Coefficients (MFCC) [155] to extract features from target radiated noise. Zhu et al. [29] focused on shipwreck recognition in side-scan sonar images. Their method extracts texture features, such as fractal dimension and multifractal width, alongside grayscale features, and applies Independent Component Analysis for

feature selection. In this framework, a Gentle AdaBoost model was employed to construct the recognition system, while nonlinear matching models facilitated rapid detection and diffusion mapping for precise localization.

Furthermore, in recent years, generative AI has been increasingly employed to address data scarcity in underwater acoustic target recognition. Yao et al. [156] introduced a novel method integrating data augmentation techniques with residual convolutional neural networks. This approach learns the latent distribution of target features from noise distributions to generate realistic synthetic samples. Similarly, Du et al. [157] proposed Multi-Discriminator Generative Adversarial Networks (MD-GANs). In this framework, the collaboration among multiple discriminators drives the generator to produce high-quality samples with enhanced deceptiveness.

2) *Deep Reinforcement Learning for Perception Optimization:* Reinforcement Learning optimizes agent policies by maximizing long-term accumulated rewards through interaction with the environment. Unlike traditional methods, DRL does not require precise prior knowledge of the environment. Instead, it demonstrates robust adaptability to dynamic changes, making it particularly suitable for perception tasks within complex underwater environments.

Existing research has explored two primary directions: “wake-up strategies” and “collaborative scheduling.” Zheng et al. [26] addressed the optimal wake-up strategy using Deep Q-Networks (DQN) and incorporated Long Short-Term Memory (LSTM) networks to accurately estimate network traffic characteristics, thereby effectively supporting strategy selection. In a related study regarding passive tracking, Su et al. [27] focused on energy-efficient scheduling within the perception layer of Underwater Wireless Sensor Networks (UWSNs). They constructed reward functions using simulated data to evaluate tracking performance based on the deviation between simulated and actual measurements. This approach effectively addressed the challenge of obtaining real trajectories for non-cooperative targets.

Regarding collaborative sensing in large-scale autonomous underwater vehicle (AUV) swarms, Zhu et al. [24] proposed advantage attention mechanisms and advantage resampling methods. Similarly, Xu et al. [25] enabled high-quality perception-action patterns without relying on explicit reward functions. Building upon these foundations, Wang et al. [158] extended the approach to multi-target tracking scenarios by integrating fuzzy logic and rule-based expert systems for AUV formation classification. This integration achieved optimal matching between multiple targets and AUV perception resources. Furthermore, Yang et al. [159] developed a method relying solely on local sensor data for autonomous decision-making, eliminating the need for real-time inter-AUV communication.

3) *Comparison and Analysis of Performance Metrics:*

Regarding Deep Neural Network-based methods, studies that do not employ data augmentation [28], [29] demonstrate accuracies ranging from 90% to 97%. These approaches rely on feature engineering, such as fractal features and LOFAR spectra, rendering them suitable for scenarios with sufficient sample availability. Conversely, methods incorporating data

TABLE VI
PERFORMANCE COMPARISONS OF PERCEPTION LAYER METHODS

Paper	Performance Metrics		
	Tracking Accuracy	System Energy Consumption	Algorithm Convergence Speed
[27]	Significantly improved vs. random/fixed strategies	Significantly reduced vs. original EACDS (42.86% duty cycle)	Iterations reduced by 30% vs. non-LSTM scheme
[26]	Improved by $\geq 20\%$ vs. DP and GA	Reduced by 10% via sensor scheduling	$40 \times$ faster than basic DQN
[159]	Highest success rate and smallest deviation (double-AUV)	Superior efficiency vs. centralized execution	Improved by 25% vs. distributed training
[25]	Avg. min distance: 13.02m (near expert level)	No additional consumption from "danger time" in dense obstacles	Converges in <80 iterations (small-scale scenarios)
[24]	97.43% (2D) / 95.28% (3D)	Stability significantly better than traditional algorithms	50% faster than MADDPG and MAAC
[158]	94.23% (no interference) / 99.90% (with interference)	Excellent scalability; non-linear increase in large swarms	Improved by 40% via reward reshaping

* This table is used to compare machine learning-driven methods for optimizing the perception layer, selecting three metrics: tracking accuracy, system energy consumption, and algorithm convergence speed.

augmentation [156] achieve a peak accuracy of 96.37%. This performance is attained by utilizing DCGANs to generate synthetic samples, thereby addressing the challenge of limited underwater data. Furthermore, the integration of the ResNet18 residual structure mitigates gradient vanishing and enhances feature extraction capabilities.

For RL-based methods, three core metrics-tracking accuracy, system energy consumption, and algorithm convergence speed-were selected to compare the performance of six studies. These metrics serve as critical evaluation dimensions for DRL methods within underwater acoustic target tracking, and all cited works provide experimental validation or supporting evidence. The comparative results are presented in Table VI.

C. Processing Layer

Optimization of the processing layer in underwater acoustic target tracking is typically approached from two complementary dimensions: the employment of DNN technologies to enhance the robustness and accuracy of state estimation, and the adoption of DRL techniques to realize dynamic optimization of collaborative decision-making and resource allocation.

The former approach learns temporal associations directly from the perception layer's feature data, making it well-suited for handling nonlinearities, high-dimensional noise, and multipath propagation inherent in complex underwater acoustic environments [160]. The latter leverages real-time feedback on tracking accuracy within the cognitive network to learn optimal strategies for task allocation, bandwidth scheduling, power control, and multi-node data fusion. Consequently, DRL achieves a critical balance between tracking accuracy and energy consumption [161].

1) *Deep Neural Networks for State Estimation Optimization*: Traditional filtering models rely on recursive updates to facilitate continuous tracking; however, their accuracy depends critically on the precise specification of noise characteristics and motion model parameters. Inaccurate parameter selection

often results in accumulated estimation bias, significantly degrading tracking performance. The adaptive representation capabilities of deep learning can be leveraged to optimize these key parameters-either online or offline-thereby enhancing overall tracking quality [162]. For instance, Xu et al. [22] proposed EKFNet, a model that retains the core EKF framework comprising state prediction, measurement prediction, and state updates. Uniquely, it utilizes backpropagation through time to learn process noise and measurement noise covariance parameters in an end-to-end manner, effectively eliminating the need for manual tuning.

Furthermore, traditional methods are highly sensitive to initial mean squared error, where improper initialization may induce tracking divergence. To address this challenge, Hou et al. [23] developed an attention-based deep convolutional neural network to robustly estimate initial noise covariance. By embedding this estimation into a variational Bayesian EKF framework, the approach effectively mitigates the impact of initialization uncertainty on the stability and accuracy of subsequent DOA tracking.

2) *Deep Reinforcement Learning for Collaborative Mechanism Optimization*: Even with reliable state estimation, the processing layer necessitates intelligent optimization of network-level collaboration and resource scheduling. Software-Defined Networking (SDN), characterized by the decoupling of the control plane from the data plane, provides a programmable and scalable system architecture [163]. As illustrated in Figure 6b, this architecture distinguishes itself from traditional distributed or centralized network configurations.

Building on this foundation, Lin et al. [20] and Casas-Velasco et al. [21] proposed an SDN-based underwater wireless network architecture for AUVs. This system integrates software-defined beacons, hierarchical localization, collaborative control, and a hybrid data transmission framework. Consequently, the architecture facilitates information

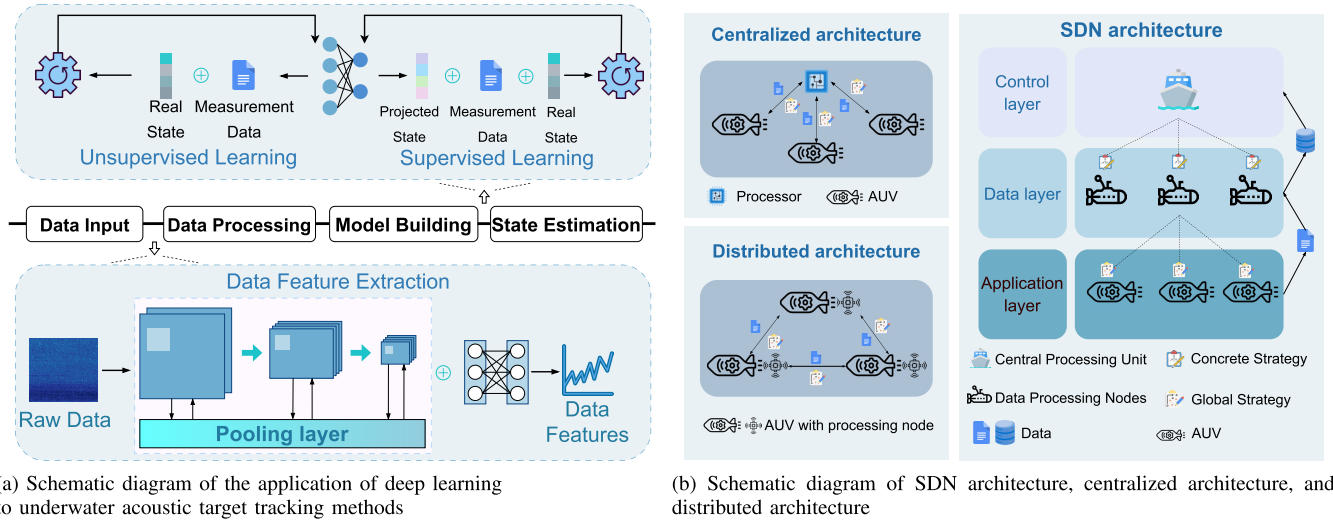


Fig. 6. The applications of machine learning technology in the field of cognitive underwater acoustic target tracking. a) shows the application of deep learning (unsupervised learning and supervised learning) in state estimation models for underwater acoustic target tracking, as well as the application of deep learning in feature extraction from target tracking measurement data. b) demonstrates the application of reinforcement learning strategies combined with SDN architecture in underwater acoustic target tracking, with strategy generation and data processing decoupled through the control layer and data layer.

synchronization, precise node localization, multi-AUV coordination, and intelligent data scheduling. Furthermore, beyond SDN-specific scenarios, routing protocols based on reinforcement learning [18], [19] demonstrate significant potential for UWSNs.

3) Comparison and Analysis of Performance Metrics:

Three core metrics-tracking accuracy, system energy consumption, and robustness-were selected to encompass the essential requirements of accuracy, energy efficiency, and environmental adaptability in underwater acoustic target tracking. All referenced studies validated these metrics through simulation or field trials, thereby facilitating direct comparative analysis. The following conclusions are drawn:

- **Tracking accuracy:** DNN-based methods [22], [23] exhibit superior performance in scenarios involving maneuvering targets and real-world ocean environments. Conversely, DRL-based methods [19], [21] demonstrate distinct advantages regarding Position Detection Rate (PDR) within multi-node collaboration and sparse coverage scenarios.
- **System energy consumption:** DRL-based methods significantly reduce redundant overhead and demonstrate substantial energy conservation through policy optimization. Meanwhile, DNN-based methods attain moderate efficiency by precisely estimating parameters to minimize energy wastage.
- **Robustness:** DNN-based methods display robust adaptability to model deviations and parameter disturbances. In contrast, DRL-based methods offer enhanced fault tolerance in scenarios characterized by dynamic network topologies and sparse node distribution.

D. Cross-Layer Optimization

In cognitive networks designed for underwater acoustic target tracking, significant interdependencies exist among the

target, perception, and processing layers. DNNs facilitate efficient cross-layer information transmission through end-to-end feature learning and model sharing. Complementarily, DRL leverages environmental interaction and closed-loop feedback mechanisms to enable the dynamic adaptation of cross-layer strategies, thereby enhancing overall system performance. The following analysis examines the optimization mechanisms and representative research cases associated with three typical cross-layer combinations.

1) *Target Layer-Perception Layer:* In the collaborative optimization of the target and perception layers, DNNs facilitate multi-target signal association while simultaneously extracting and enhancing acoustic signal features. Consequently, this approach improves both association accuracy and perception quality within a unified framework.

For instance, to address perception requirements in deep-sea environments, Liu et al. [152] constructed a three-dimensional feature tensor utilizing the real and imaginary components of the signal separation matrix. To mitigate performance degradation caused by data scarcity, Wang et al. [33] implemented data augmentation techniques involving the shifting of signal integration time and azimuth. This strategy enhanced the generalization capability of perception features, yielding superior target extraction performance. Furthermore, Wang et al. [32] proposed a perception method based on multi-beam LOFAR spectra to address the concealment requirements inherent in passive tracking.

2) *Target Layer-Processing Layer:* Cross-layer optimization between the target and processing layers primarily utilizes multi-task learning frameworks to establish a direct mapping from target motion prediction to state estimation. This methodology facilitates the dynamic adjustment of inference resource allocation within the processing layer based on prediction accuracy. Simultaneously, it enables the continuous correction

TABLE VII
COMPARISONS BETWEEN TRADITIONAL METHODS AND MACHINE LEARNING-DRIVEN METHODS

Comparison Dimensions	Traditional Methods		ML-based Methods	
	Advantages	Disadvantages	Advantages	Disadvantages
Adaptability	Clear logic mature engineering	Fixed models prone to mismatch	Data-driven strong generalization	Requires large labeled data
Robustness	Solid theory stable	Bias accumulation weak anti-interference	Suppresses noise and interference	Overfitting in extreme environments
Computational Cost	No training low pressure	High complexity reduced real-time performance	Reduces manual engineering	High resources for training

* The table compares the advantages and disadvantages of traditional methods and machine learning-driven methods from three aspects: adaptability, robustness, and computational cost.

of motion hypotheses in the target layer, guided by feedback from the processing layer.

For instance, Niu et al. [30] proposed a two-stage ResNet50 processing framework. In the first stage, coarse grid classification with 5 km intervals determines the target's approximate region. Subsequently, the second stage employs fine-grained models-with resolutions of 0.1 km or 2 m-within the identified region to precisely estimate target range and depth.

Huang et al. [31] employed a Time Delay Neural Network (TDNN) to construct a regression processing model, leveraging its temporal feature modeling capabilities to map perception-layer feature vectors directly to target range and depth. Furthermore, Wang et al. [33] introduced the DCNN-KFAT hybrid processing framework. In this approach, a KF first predicts target azimuth and distance to narrow the search space. A pre-trained DCNN then analyzes azimuth-weighted time-frequency maps to determine the target category and precise location. Finally, the KF corrects initial state biases and updates the covariance matrix.

3) *Perception Layer-Processing Layer*: Optimization interaction between the perception and processing layers focuses on the effective transformation of low-level perception features into high-level state estimates, as well as the efficient allocation of resources.

Yao et al. [28] proposed a processing framework utilizing data augmentation and ResNet18. In this approach, Mel Frequency Cepstral Coefficient (MFCC) features are extracted at the perception layer. Subsequently, a residual connection structure is employed to mitigate the gradient vanishing problem inherent in deep networks, thereby enhancing the deep representation of target acoustic features.

Regarding resource scheduling, Zheng et al. [27] introduced an end-to-end sensor scheduling scheme based on Double Deep Q-Networks (D3QN). By constructing reward functions to balance tracking accuracy with energy consumption, this method eliminates the reliance on true target trajectories. Similarly, Su et al. [26] developed a processing framework that integrates Deep Reinforcement Learning with LSTM networks to address energy constraints within underwater acoustic sensor networks. Furthermore, Zhu et al. [24] proposed a Hierarchical Software-Defined Multi-AUV Reinforcement Learning (HSD-MARL) architecture. By decomposing collaborative decision-making into global training and local

execution phases, this architecture achieves cooperative optimization in multi-target tracking task allocation and trajectory updating.

E. Summary

The optimization of cognitive systems for underwater acoustic target tracking is organized around the target, perception, and processing layers of the cognitive network. Traditional methods and emerging approaches based on DNN and DRL exhibit distinct disparities in both methodological frameworks and performance outcomes. Variations in adaptability, robustness, and computational cost significantly impact their engineering applicability and tracking efficacy across diverse scenarios. The comparison results are shown in Table VII.

In summary, traditional methods are fundamentally model-driven. These approaches execute tracking tasks through filtering algorithms, data association mechanisms, and pre-determined collaborative strategies, relying heavily on hand-crafted features and prior physical assumptions.

In contrast, approaches driven by DNN and DRL operate on a data-driven paradigm. They adapt to complex underwater environments and dynamic target characteristics by leveraging automatic feature extraction, dynamic policy optimization, and cross-layer collaborative mechanisms.

V. CHALLENGES AND FUTURE AVENUES

This chapter systematically examines key challenges in underwater acoustic target tracking research, focusing on three critical aspects: **data processing, algorithm design, and practical deployment**. Furthermore, potential solutions leveraging emerging technologies-such as physics-informed machine learning and embodied intelligence-are proposed. The interrelationship between current challenges and future research directions is illustrated in Figure 7.

A. Current Challenges

1) *Limited Sharing of Data*: Research in underwater acoustic target tracking relies heavily on acoustic data collected from real marine environments. However, constraints related to security, privacy, and intellectual property often preclude the public sharing of such datasets. This limitation restricts model validation and comparative analysis, while also impeding

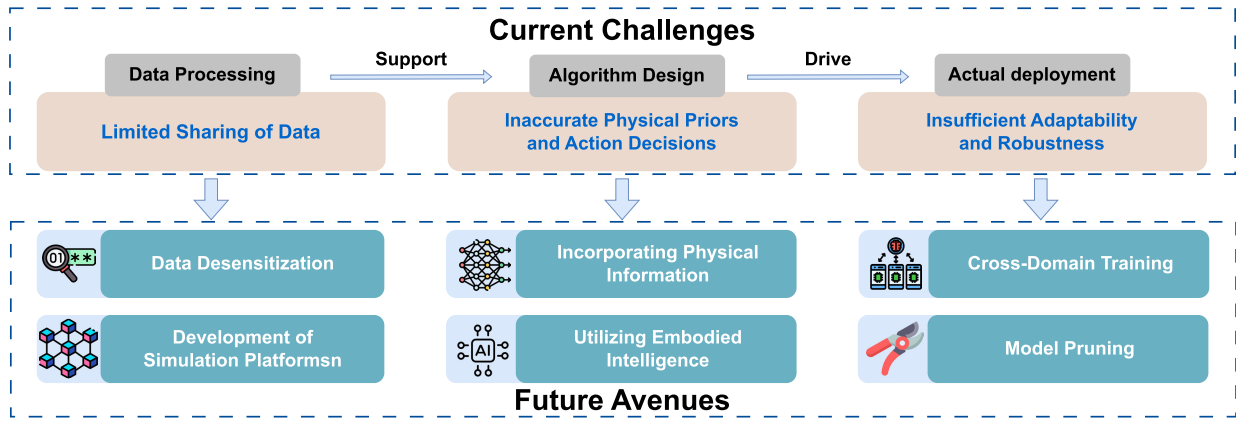


Fig. 7. Current challenges and future avenues. The upper half of the figure illustrates three current challenges: data processing, algorithm design, and actual deployment. The lower half presents corresponding future avenues by integrating techniques such as PINN, embodied intelligence, data desensitization, and model pruning.

the improvement of algorithm generalization and innovation. Consequently, data scarcity constitutes a significant bottleneck restricting further advancements in the domain.

2) *Inaccurate Physical Priors and Action Decisions*: Cognitive underwater acoustic target tracking systems typically rely on physical prior knowledge for environmental modeling and decision-making. However, the physical priors utilized in existing systems often fail to accurately represent complex real-world scenarios. Furthermore, action decision modules frequently exhibit insufficient adaptability, hindering their ability to cope with dynamic target behaviors and environmental variations.

3) *Insufficient Adaptability and Robustness*: While underwater acoustic target tracking algorithms often demonstrate promising results in controlled experimental settings, they frequently struggle with environmental adaptability and robustness in complex, dynamic real-world scenarios. Specifically, these algorithms exhibit limited generalization across diverse geographical regions, varying noise levels, and heterogeneous hardware configurations. Consequently, their susceptibility to environmental fluctuations and interference leads to significant performance degradation.

B. Future Avenues

To address the limitations in data processing, we propose exploring the following strategies:

1-1) *Data Desensitization*: The development of general standards and automated tools for underwater acoustic data desensitization is essential. For instance, integrating emerging data security methods such as federated learning and differential privacy [164] allows for local training of key models. By encrypting the transmission of critical parameters, data privacy and security can be maintained during sharing [165].

1-2) *Development of Simulation Platforms*: There is a need to integrate models across various stages of underwater acoustic target tracking to develop comprehensive simulation platforms. Drawing upon the concept of decentralization from blockchain technology, portions of model information and underwater acoustic data can be open-sourced, while private

information remains encrypted. Storing these diverse information types in a distributed manner [164] will further enhance the flexibility and security of data sharing.

To address the limitations in algorithm design, we propose exploring the following strategies:

2-1) *Incorporating Physical Information*: Leveraging Physics-Informed Neural Networks (PINNs) allows for the embedding of physical laws and environmental constraints directly into intelligent tracking algorithms [166]. This integration improves the algorithms' ability to adaptively adjust to physical priors, thereby enhancing system modeling capabilities and generalization performance within complex underwater environments [167].

2-2) *Utilizing Embodied Intelligence*: Future research should prioritize the integration of Embodied Intelligence to establish real-time perception and dynamic feedback mechanisms within cognitive systems [168]. To achieve this, studies should look beyond basic interaction to adopt advanced paradigms currently emerging in autonomous underwater platforms. These include imitation learning for autonomous strategy acquisition [169], modular frameworks based on Multimodal Large Language Models (MLLMs) for high-level reasoning [170], and end-to-end Vision-Language-Action (VLA) foundation models for direct sensor-to-control mapping [171]. Incorporating these technologies will empower cognitive systems to derive complex control policies from continuous environmental interaction, thereby significantly enhancing decision-making precision in target tracking.

To address the limitations in practical deployment, we propose exploring the following strategies:

3-1) *Cross-Domain Training*: Collecting diverse underwater acoustic datasets that encompass varying environmental characteristics is crucial. Through the design of adversarial networks and domain generalization loss functions, models can be trained to extract domain-invariant features. This approach strengthens the learning of stable, essential features, significantly improving robustness against environmental changes [172].

3-2) *Model Pruning*: Applying lightweight operations to large-scale models is necessary for efficient deployment.

Techniques such as knowledge distillation allow a “teacher” model to transfer key knowledge to a deployable “student” model. This process facilitates model compression by removing redundant connections and parameters [173], thereby reducing computational requirements and resource consumption in practical applications.

VI. CONCLUSION

This comprehensive survey introduces a novel three-dimensional classification framework for cognitive underwater acoustic target tracking networks. By categorizing recent advancements according to the three-layer architecture of this cognitive network, it systematically reviews the associated theoretical foundations and algorithmic developments.

Emphasizing the integration of ML—particularly DNN and DRL—within cognitive communications and networking, this work highlights how such methods significantly optimize each layer of the cognitive underwater acoustic target tracking networks, thereby enhancing tracking accuracy, adaptability, and intelligence.

Nevertheless, challenges persist across three critical domains: data processing, algorithm design, and practical deployment. To address these impediments, the further application of AI methodologies, including data desensitization, PINNs, and embodied intelligence, is essential.

This survey aims to guide researchers toward the development of more adaptive, trustworthy, and efficient underwater acoustic networks, ultimately accelerating the transition from traditional sensing systems to fully cognitive marine cyber-physical infrastructures.

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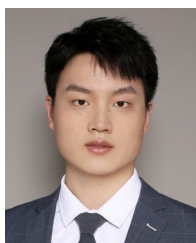
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